Relations and functions

1 Express the relation $\{(2,2),(4,4),(6,3),(7,5)\}$ in a table of values.

2 Determine whether the following statements are true or false:

a When working with a function, substituting a certain value of x into the formula gives only one value of y for that value of x.

b All relations are functions.

c Some relations are functions.

d All functions are relations.

e No functions are relations.

f No relations are functions.

g A relation always passes the vertical line test.

h The graph of every straight line is a function.

i The graph of every non-vertical straight line is a function.

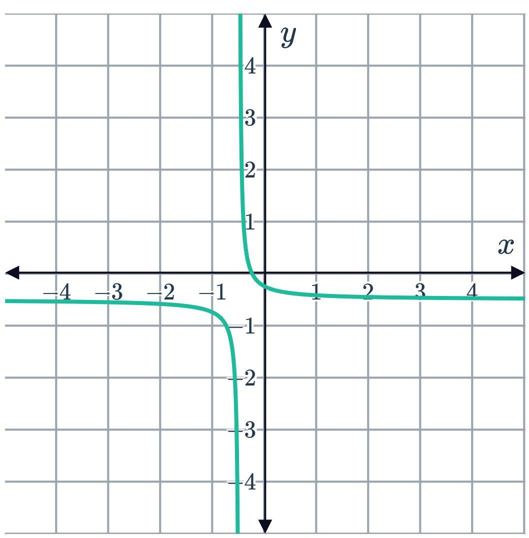
j There is no straight line graph that is a function.

k The graph of every straight line is a relation.

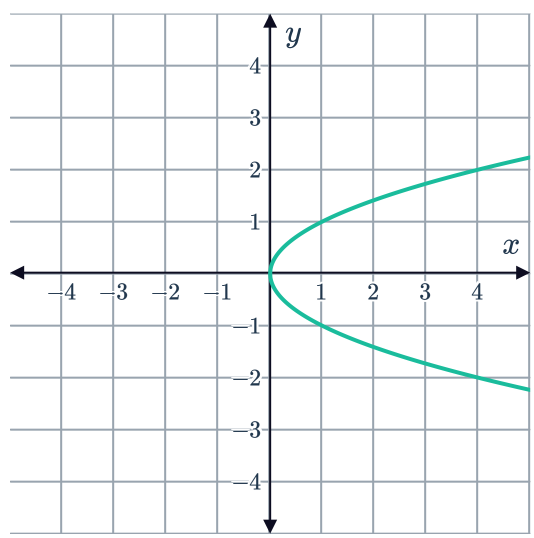
l The graph of every non-horizontal straight line is a function.

3 Determine whether each of the following graphs is a relation only or both a relation and a function:

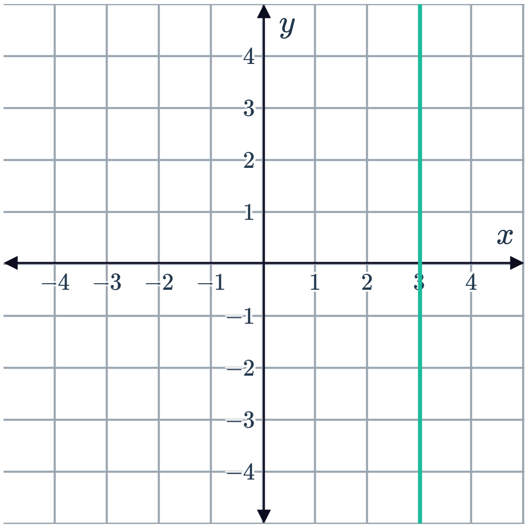
a



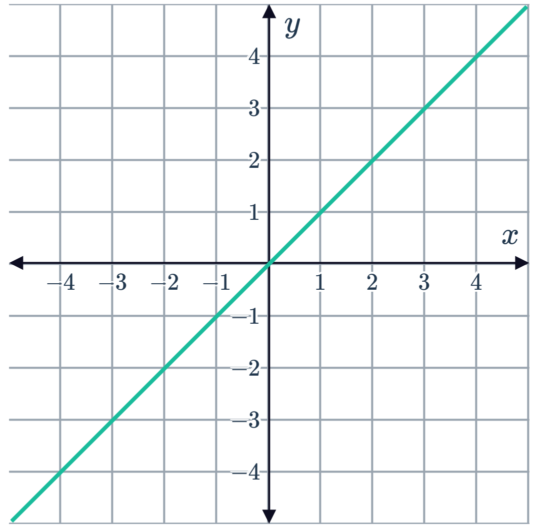
b



c

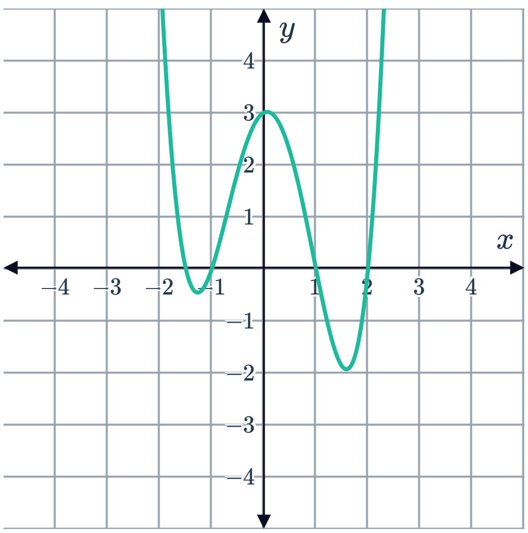


y
x
4
3
2
1
−1
−2
−3
−4



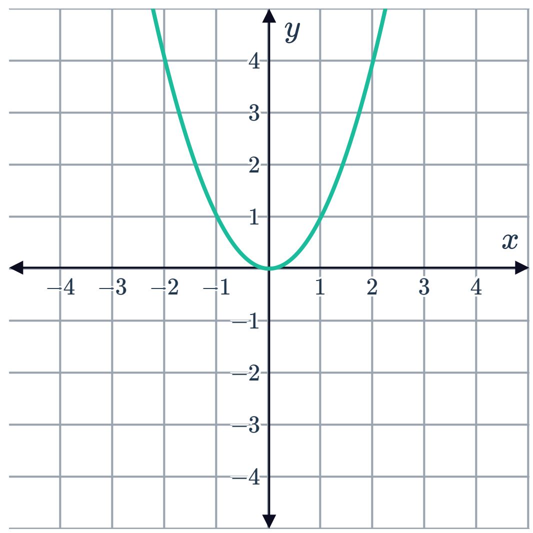
y
x
4
3
2
1
−1
−2
−3
−4

e



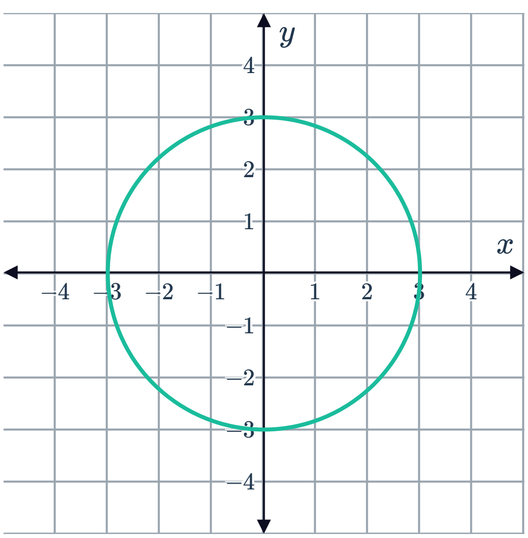
y
x
4
3
2
1
−1
−2
−3
−4

f



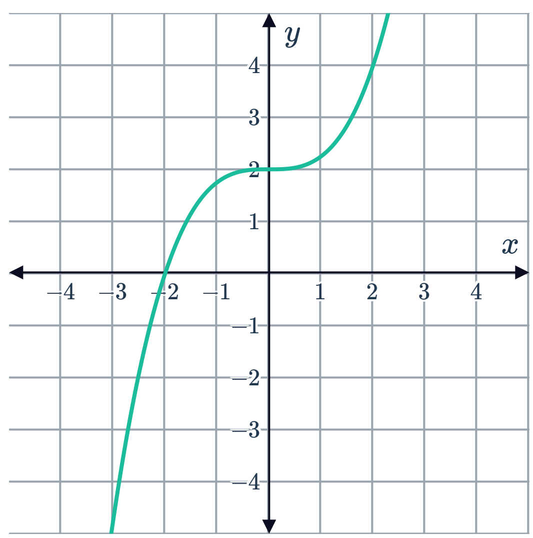
y
x
4
3
2
1
−1
−2
−3
−4

g



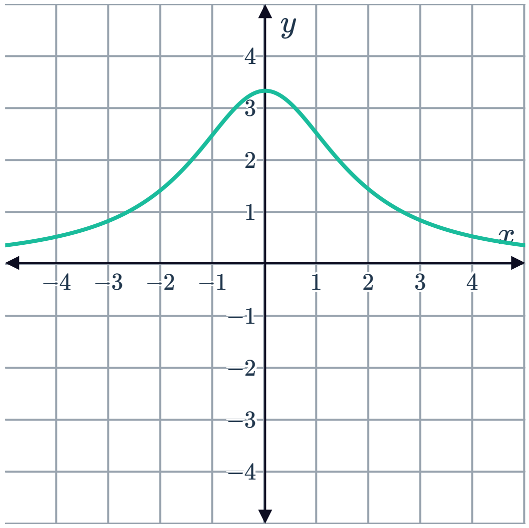
y
x
4
3
2
1
−1
−2
−3
−4

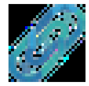
h

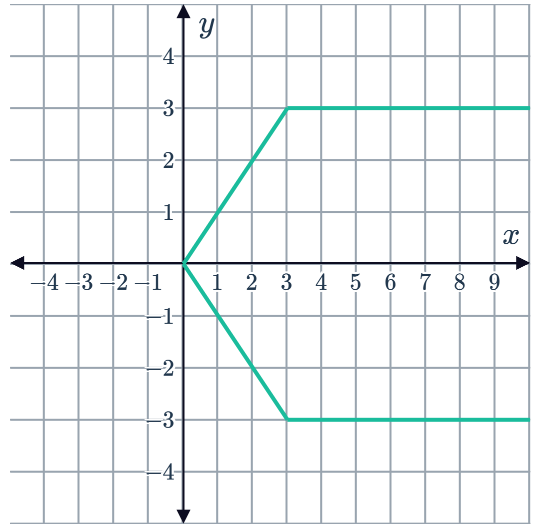


y
x
4
3
2
1
−1
−2
−3
−4

i







4 Consider the following set of points: $\{(16,14),(8,28),(-15,14),(-20,-28)\}$

a List the domain of the set of points.

b List the range of the set of points.

c State whether the points describe a function.

5 State whether the following set of points describe a function, a relation, or both:

a $\{(2,5),(7,-3),(5,2),(-4,-9)\}$

b $\{(2,5),(2,7),(-3,-4),(-9,13)\}$

c $(1,5),(1,1),(7,-2),(-5,-10)$

d $(1,5),(7,-2),(-5,-10),(13,-13)$

e $(1,5),(1,7),(-2,-5),(-5,-10)$

6 A relation is defined as follows:
$y=1$ if x is positive and $y=-1$ if x is 0 or negative.

a Complete the following table:

x	-4	-3	-2	-1	0	1	2	3	4
y									

b Plot the points on the coordinate plane.

c State whether the relation is a function.

7 Determine whether the pairs of values in th[redacted]es represent a function between x and y:

a

x	-4	-3	-2	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	-1	-2	-3	-4

b

x	0	1	4	8	9	12	16	18	20
y	0	1	2	$2\sqrt{2}$	3	$2\sqrt{3}$	4	$3\sqrt{2}$	$2\sqrt{5}$

c

x	-9	-7	-6	-5	-3	-2	3	5	10
y	10	10	10	10	10	10	10	10	10

d

x	-8	-7	-6	-3	2	7	9	9	10
y	8	13	-18	-16	-15	-2	-4	11	-9

8 Consider these ordered pairs:
$\{(-9,-5),(-5,-10),(-5,-4),(-3,7),(-2,-4),(-1,1)\}$

a Plot the ordered pairs on the number plane.

b State whether the relation is a function.

c Identify the ordered pair needed to be removed from the set so that the remaining ordered pairs represent a function.

9 Find the possible values of k such that each of the following relations does not represent a function.

a $\{(2,1),(7,k),(5,8),(3,9)\}$

b $\{(6,2),(8,5),(1,7),(k,4)\}$

Function notation

10 For the following functions, determine the output produced by the indicated input value:

a $f\left(x\right)=8x+6; x=5$

b $f\left(x\right)=-9x^2-8x-4; x=-5$

c $g\left(t\right)=-\dfrac{2}{t}+8; t=-8$

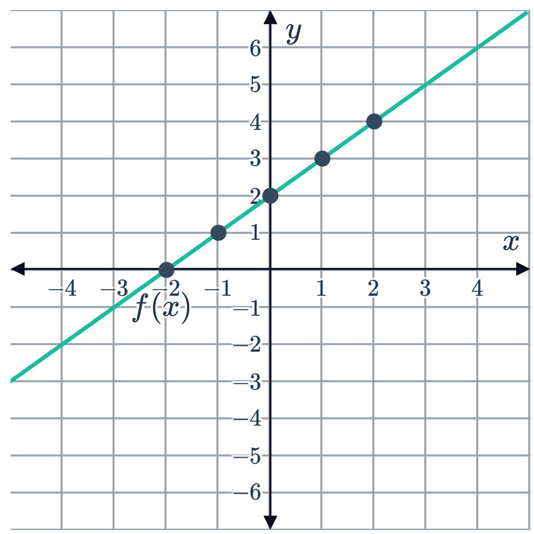
d $h\left(t\right)=9^t+6; t=2$

11 Rewrite the following statements using function notation:

a Suppose that $(7,-6)$ is an ordered pair that satisfies the function g.

b In the equation $x-4y=8$, y is a function of x.

12 Consider the following graph of the function $f\left(x\right)$:



a Find $f\left(0\right)$.

b Find $f\left(-2\right)$.

c Find the value of x such that $f\left(x\right)=3$

13 Consider the function $h\left(x\right)=-x^2+6x-6$.

a For each input value of x, state the maximum number of distinct output values $h(x)$ can produce.

b Is $h\left(x\right)$ a function?

14 If $f\left(x\right)=4x+4$, find:

a $f\left(2\right)$

b $f\left(-5\right)$

15 If $f\left(x\right)=3x-1$, find:

16 If $g(x) = \dfrac{7x}{4}$, find:

a $g(5)$

b $g(-4)$

17 If $f(x) = \dfrac{2x-3}{5}$, find:

a $f(4)$

b $f(-5)$

18 Consider the function $f(x) = 4 + x^3$.

a Evaluate $f(4)$

b Evaluate $f(-2)$

19 Consider the function $f(x) = 2x^2 - 2x + 5$. Evaluate $f\left(\dfrac{1}{2}\right)$.

20 If $f(t) = \dfrac{t^3 + 27}{t^2 + 9}$, evaluate the following:

a $f(-3)$

b $f(3)$

c $f(4)$

21 Consider the function $f(x) = x^2 - 49$.

a Find:

i $f(1)$

ii $f(8)$

iii $f(0)$

b If $f(x) = 12$, what are the possible values for x, rounded to two decimal places?

22 Consider the function $f(x) = x^2 + 8x$.

a Form an expression for $f(a)$.

b Form an expression for $f(b)$.

23 Consider the function $f(x) = 2x^3 + 3x^2 - 4$.

a Evaluate $f(0)$.

b Evaluate $f\left(\dfrac{1}{4}\right)$.

24 If $j(x) = 3^x - 3^{-x}$, find the following, rounding your answers to two decimal places where necessary:

a $j(0)$ b $j(1)$ c $j(4)$

25 A function is defined as $f(x) = 2 + 3x - x^3$. Evaluate $f(-2) + f(4)$.

26 If $A(x) = x^2 + 1$ and $Q(x) = x^2 + 9x$, evaluate:

a $A(5)$ b $Q(4)$ c $A(3) + Q(2)$

27 If $m(x) = \sqrt{12^2 - x^2}$, find:

a $m(4)$ b $m(0)$ c $m(9)$ d $m\left(\sqrt{2}\right)$

28 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$. b Form an expression for $p(m)$.

29 Consider the function $f(x) = x^3 - 8x + 7$.

a Evaluate $f(4)$.

b Evaluate $f(-4)$.

c Form an expression for $f(a)$.

d Form an expression for $f(b)$.

e Form an expression for $f(a+b)$.

f Is $f(a) + f(b) = f(a+b)$ for all values of a and b?

30 Consider each of the following equations.

i Rewrite the equation using function notation, $f(x)$.

ii Find the value of $f(3)$.

a $x + 3y = 6$ b $-6x + 5y = 7$

c $y + 6x^2 = 3 - x$

31 Consider each of the following equations.

i Rewrite the equation using function notation $f(x)$.

ii Find the value of $f(2)$.

a $5x - 3y = 7$

b $y - 4x^2 = 5 + x$

32 Consider the equation $x - 4y = 8$. Assume that y is a function of x.

a Rewrite the equation using function notation $f(x)$.

b Find the value of $f(12)$.

33 Consider $f(x) = x + 3$ for the domain $\{-5, -4, 0, 1\}$.

a Complete the table of values:

x	-5	-4	0	1
$f(x)$				

b Plot the points on the coordinate plane.

34 Consider $f(x) = -2x$ for the domain $\{-1, 0, 1, 2\}$.

a Complete the table of values:

x	-1	0	1	2
$f(x)$				

b Plot the points on the coordinate plane.

35 Consider $f(x) = (x + 2)^2$ for the domain $\{-2, -1, 0, 1, 2\}$.

a Complete the table of values:

x	-2	-1	0	1	2
$f(x)$					

b Plot the points on the coordinate plane.

36 Consider $f\left(x\right) = x^2 + 4$ for the domain $\{-3$ [redacted] $, 1, 3\}$.

a Complete the table of values:

x	-3	-1	0	1	3
$f\left(x\right)$					

b Plot the points on the coordinate plane.

37 Consider $f\left(x\right) = \left|x\right| + 4$ for the domain $\{-4, -2, 0, 2, 4\}$.

a Complete the table of values:

x	-4	-2	0	2	4
$f\left(x\right)$					

b Plot the points on the coordinate plane.

Applications

38 Dodgi Mobile charges $\$2.90$ a minute plus a connection fee of $\$0.60$ for an international call.

a Complete the table.

Call length (minutes)	International Call cost
1	
2	
3	
4	
5	

b State whether the relation is a function.

39 A particular store is offering one free T-shirt [illegible]very two T-shirts purchased, where each T-shirt costs $\$19$.

a Complete the table.

Number of T-shirts	Total Cost (dollars)
1	
2	
3	
4	
5	

b State whether the relation is a function.

40 Tyrah makes scarfs to sell at the market. It costs her $\$2$ to produce each one, and she sells them for $\$5$.

a Consider when 1, 2, 3, 4 and 5 scarfs are sold. Graph the points representing the relation between the number of scarfs she manages to sell and her total profit.

b State whether the relation is a function.